

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0940

Hubbardstown 345 kV
May 20, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0940. The Interconnection Customer proposes a 1339.3 MW natural gas combined cycle facility interconnecting at the Hubbardstown 345 kV substation in the Kentucky area. The request was received on February 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 366 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$56,622,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Hubbardstown-Hyampom 138kV: 130.1% loading (323 MVA) following the outage of the Hubbardstown-Tristate 345kV Line; DFAX = 9.9%.

Siggen-Hyampom 138kV: 122.4% loading (280 MVA) following the outage of the Hubbardstown Substation-Tristate 345kV Line; DFAX = 8.6%.

Hubbardstown-Tristate Substation 345kV: 116.4% loading (1,167 MVA) following the outage of the International Paper-Hanging Rock 500kV L; DFAX = 42.3%.

The Aspen-Lowmoor 345kV is overloaded to 109.2% of its 1,321 MVA emergency rating for the loss of the Bath County-Longview2 345kV line. The project contributes 6.7% DFAX (89.6 MW) to this overload.

Northwest Beloit 230 kV/1 kV XFMR: 107.5% loading (415 MVA) following the outage of the Tristate-Valemount 345kV; DFAX = 13.1%.

Sporn-Beck Road 230kV: 106.1% loading (619 MVA) following the outage of the Sporn-Beck Rd 345kV line; DFAX = 6.2%.

Lowmoor-Hinton Ap 345kV: 105.4% loading (1,182 MVA) following the outage of the McClain Energy Facility-Vaseaux Lake 138kV; DFAX = 8.9%.

The North Pt 345/138 kV Transformer is overloaded to 105.0% of its 478 MVA emergency rating for the loss of the Dewey-Dowell 138kV. The project contributes 5.9% DFAX (79.3 MW) to this overload.

Neptune-Kettle Creek Primary 138kV Line: 104.0% loading (362 MVA) following the outage of the Hubbardstown-Tristate 345kV; DFAX = 5.7%.

Mt Vernon-Kettle Crk Primary 138kV: 103.4% loading (377 MVA) following the outage of the Hubbardstown Substation-Tristate 345kV; DFAX = 5.7%.

The Pocket-Mt Vernon 138kV is overloaded to 103.0% of its 387 MVA emergency rating for the loss of the Hubbardstown Substation-Tristate Substation 345kV line. The project contributes 5.7% DFAX (76.3 MW) to this overload.

For the loss of Hubbardstown-Tristate 345kV, the Pocket-Tillamook 138kV loads to 102.6% of its 398 MVA rating. Project DFAX: 5.7%.

The Dewey Substation-Dowell 138kV Line is overloaded to 102.4% of its 412 MVA emergency rating for the loss of the Hubbardstown-Tristate 345kV. The project contributes 5.7% DFAX (76.5 MW) to this overload.

For the loss of Tristate-Hubbardstown 345kV, the Dewey-Tillamook 138kV line loads to 102.2% of its 408 MVA rating. Project DFAX: 5.7%.

For the loss of Tristate-Valemount 345kV Line, the Northwest Beloit 345/1 kV XFMR loads to 101.4% of its 440 MVA rating. Project DFAX: 13.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hubbardstown-Hyampom 138kV	Rebuild line with higher-capacity conductor	\$3,110,000	49% (\$1,521,000)
Siggen-Hyampom 138kV	Rebuild line with higher-capacity conductor	\$4,041,000	46% (\$1,850,000)
Hubbardstown-Tristate Substation 345kV	Rebuild line with higher-capacity conductor	\$36,013,000	38% (\$13,822,000)
Aspen-Lowmoor 345kV Line	Reconductor line	\$28,083,000	10% (\$2,900,000)
Northwest Beloit 230 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$7,800,000	71% (\$5,523,000)
Beck Rd-Sporn 230kV	Reconductor line	\$2,022,000	22% (\$446,000)
Hinton Ap Substation-Lowmoor 345kV	Reconductor line	\$76,003,000	9% (\$6,918,000)
North Pt 345/138 kV Transformer	Replace transformer with higher-rated unit	\$23,450,000	23% (\$5,440,000)
Neptune-Kettle Crk Primary 138kV	Reconductor line	\$7,794,000	41% (\$3,179,000)
Mt Vernon-Kettle Crk Primary 138kV	Reconductor line	\$908,000	41% (\$370,000)
Pocket-Mt Vernon 138kV	Reconductor line	\$1,797,000	41% (\$733,000)
Pocket-Tillamook 138kV	Reconductor line	\$2,435,000	41% (\$994,000)
Dewey-Dowell 138kV	Reconductor line	\$3,376,000	41% (\$1,379,000)
Tilamook-Dewey 138kV line	Reconductor line	\$4,701,000	41% (\$1,919,000)
Northwest Beloit 345/1kV XFMR	Replace transformer with higher-rated unit	\$13,597,000	71% (\$9,628,000)
	Total allocated to project		\$56,622,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.